Title: From 2D to 3D: Construction of a 3D Parametric Model for Detection of Dental Roots Shape and Position from a Panoramic Radiograph

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Date: February 14, 2026
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2. Background of the Project

Introduction and Context

Panoramic radiographs (PANs) are the cornerstone of dental imaging due to their low cost, minimal radiation, and broad view of the oral cavity. However, they provide only 2D information, critically lacking data on the bucco-lingual dimension of dental roots. This poses a significant risk during treatment planning for orthodontics, extractions, or implant placement, where the precise 3D position of roots is essential to avoid complications like root perforation or resorption (Mazzotta et al., 2013).

This project is motivated by an internship at L'Hôpital Général de Yaoundé, where PANs are the primary diagnostic tool. A key observation was the frequent poor quality of these images due to patient movement, improper positioning, or equipment limitations. This results in distorted, low-contrast images, further obscuring critical root details and making accurate diagnosis and treatment planning a challenge.

Literature Snapshot and Gaps

Obtaining 3D root information typically requires Cone Beam Computed Tomography (CBCT), which offers high accuracy (Baumgaertel et al., 2009) but at the cost of higher radiation, limiting its routine use (SEDENTEXCT, 2011). Computational alternatives exist. Mazzotta et al. (2013) showed it is possible to build a 3D parametric tooth

model using manual measurements from a single PAN. While promising, this method is labor-intensive, produces simplified root shapes, and cannot account for image distortions. Conversely, modern deep learning models like 3D-R2N2 (Choy et al., 2016) achieve impressive 3D reconstructions but are trained on ideal, multi-view datasets, not the single, noisy images found in a typical clinic.

The identified gap is the lack of a robust, automated method that can extract reliable 3D root information from the type of single-view, often suboptimal panoramic images seen at L'Hôpital Général de Yaoundé.

3. Problem Statement

At L'Hôpital Général de Yaoundé, critical dental treatment plans are formulated using 2D panoramic radiographs that often suffer from poor quality (distortion, low contrast). These images provide no information on the bucco-lingual position of roots, creating a tangible risk of iatrogenic complications during treatment. This project will address the problem of how to reliably reconstruct the 3D shape and position of dental roots from such standard, and often suboptimal, 2D panoramic images.

4. Objectives

General Objective

To develop and validate an automated framework for constructing a 3D parametric model of dental roots from a single 2D panoramic radiograph, robust to varying image qualities.

Specific Objectives

1. **To curate and pre-process a clinical dataset** of panoramic radiographs from the hospital, using enhancement techniques (e.g., CLAHE) to improve image quality and annotating key root landmarks (CEJ, apices).
2. **To design a deep learning-based feature extractor** (e.g., using EfficientNet-B0) to automatically measure critical root parameters (length, width, angulation) from the pre-processed images.

3.**To develop a parametric 3D reconstruction module** that uses these measurements to generate a patient-specific 3D mesh model of the tooth root.
4.**To validate the model's accuracy** by comparing generated 3D roots against ground-truth segmentations from available CBCT scans using metrics like Mean Absolute Error (MAE) and Dice Similarity Coefficient (DSC).

Expected Outcomes

•An annotated dataset of paired PANs and 3D root models.
•An automated software pipeline to generate a 3D root model from a PAN.
•A validated tool to assist clinicians at L'Hôpital Général de Yaoundé in safer treatment planning without the need for a CBCT.

5. Proposed Methodology

Brief Description of Approach

The project will follow a four-stage methodology over six months.

1.**Data Acquisition & Pre-processing:** Collect anonymized PANs and corresponding CBCTs from the hospital. Develop a pipeline to enhance PAN quality (contrast, noise reduction) and manually annotate key root landmarks. Segment 3D root models from CBCTs to serve as ground truth.
2.**Feature Extraction (Deep Learning):** Train a CNN (e.g., EfficientNet-B0) on the enhanced PANs to automatically predict key root measurements (length, width). Data augmentation will be used to make the model robust to poor image quality.
3.**3D Parametric Model Construction:** Build a parametric tooth model using a 3D library (e.g., Python's trimesh). The CNN's predictions will be used to automatically scale and deform this generic model into a patient-specific 3D root.
4.**Validation:** Compare the generated 3D models against the ground-truth CBCT segmentations using quantitative metrics (MAE, RMSE, DSC) and 3D comparison software.

Resources Required

•**Software:** Python (TensorFlow/PyTorch, OpenCV, Trimesh), 3D Slicer.
•**Hardware:** Workstation with a dedicated GPU.
•**Data:** Anonymized patient data from L'Hôpital Général de Yaoundé.

Gantt Chart (6-Month Timeline)

Task	Description	M1	M2	M3	M4	M5	M6
T1	Ethics, Data Collection & Anonymization	■■■■					
T2	Image Pre-processing & Annotation		■■■■	■■■■			
T3	Deep Learning Model Development				■■■■		
T4	3D Parametric Model Integration					■■■■	
T5	Validation & Error Analysis						■■■■
T6	Final Report & Presentation						■■■■

6. References

Baumgaertel, S., et al. (2009). Reliability and accuracy of cone-beam computed tomography dental measurements. American Journal of Orthodontics and Dentofacial Orthopedics, 136(1), 19-25.

Choy, C. B., et al. (2016). 3D-r2n2: A unified approach for single and multi-view 3D object reconstruction. Computer Vision–ECCV 2016, 628-644.

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